

The background of the slide is a photograph of an offshore oil drilling rig and a support vessel on the ocean at sunset. The rig is a large, dark structure with a tall derrick, silhouetted against a sky with orange and yellow clouds. A support vessel is visible to the left of the rig. The title text is overlaid on the center of the image.

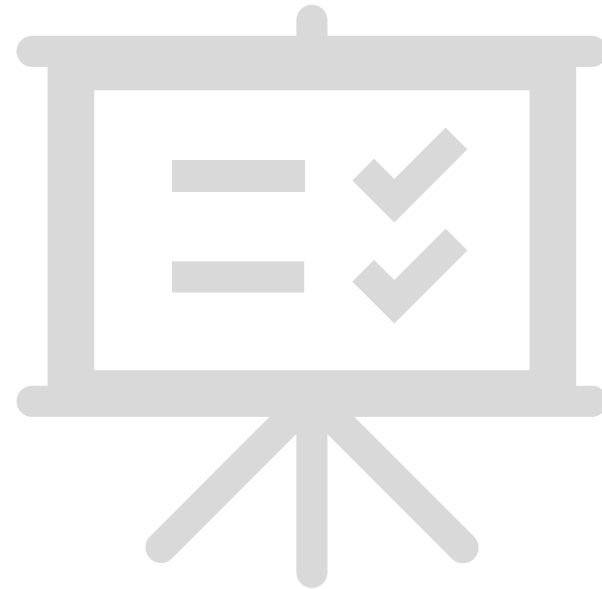
RISK MANAGEMENT FOR DRILLING PROJECT

Jakarta Drilling Society

June 3rd, 2026

Outline

- Risk Management Methodology
- Definitions
- Hierarchy of Control
- Drilling Hazard Identification
- Risk Management
- Defining Criteria



Methodology/ Tools

Method/ Tools	Purpose (as per procedure)	Application Stage
HAZID (Hazard Identification)	Systematically identify hazards across activities, facilities, and lifecycle stages	Early project / concept / planning
HAZOP (Hazard and Operability Study)	Identify deviations from design intent and operational issues	Design / engineering phase
JSA / TRA (Job Safety Analysis / Task Risk Assessment)	Identify hazards at task level and define controls	Operational / task execution
What-if Analysis	Evaluate potential abnormal scenarios and deviations	Design / review stage
Checklist / Inspection	Identify common or known hazards using predefined checklist	Routine operation / audit
Incident Investigation	Identify hazards from past incidents and lessons learned	Post-incident / continuous improvement
Environmental Aspect Identification (EAIA)	Identify environmental impacts from activities and lifecycle perspective	All lifecycle stages

Definitions

Hazard:

The potential to cause harm, including ill-health and injury, damage to assets, products or the environment, production losses or increased liabilities.

Hazard Identification:

Process of recognizing the existing of hazard and defining its characteristics.

Consequences:

Adverse effect, harm, damage or other similar loss arising from a specified event.

Likelihood:

The qualitative or quantitative description of the probability or frequency of the consequence of a risk event occurring.

Risk:

Combination of likelihood occurrence of a hazardous event or exposure(s) and its consequences or chance of occurrence hazard exposures resulted in a consequences.

Definitions

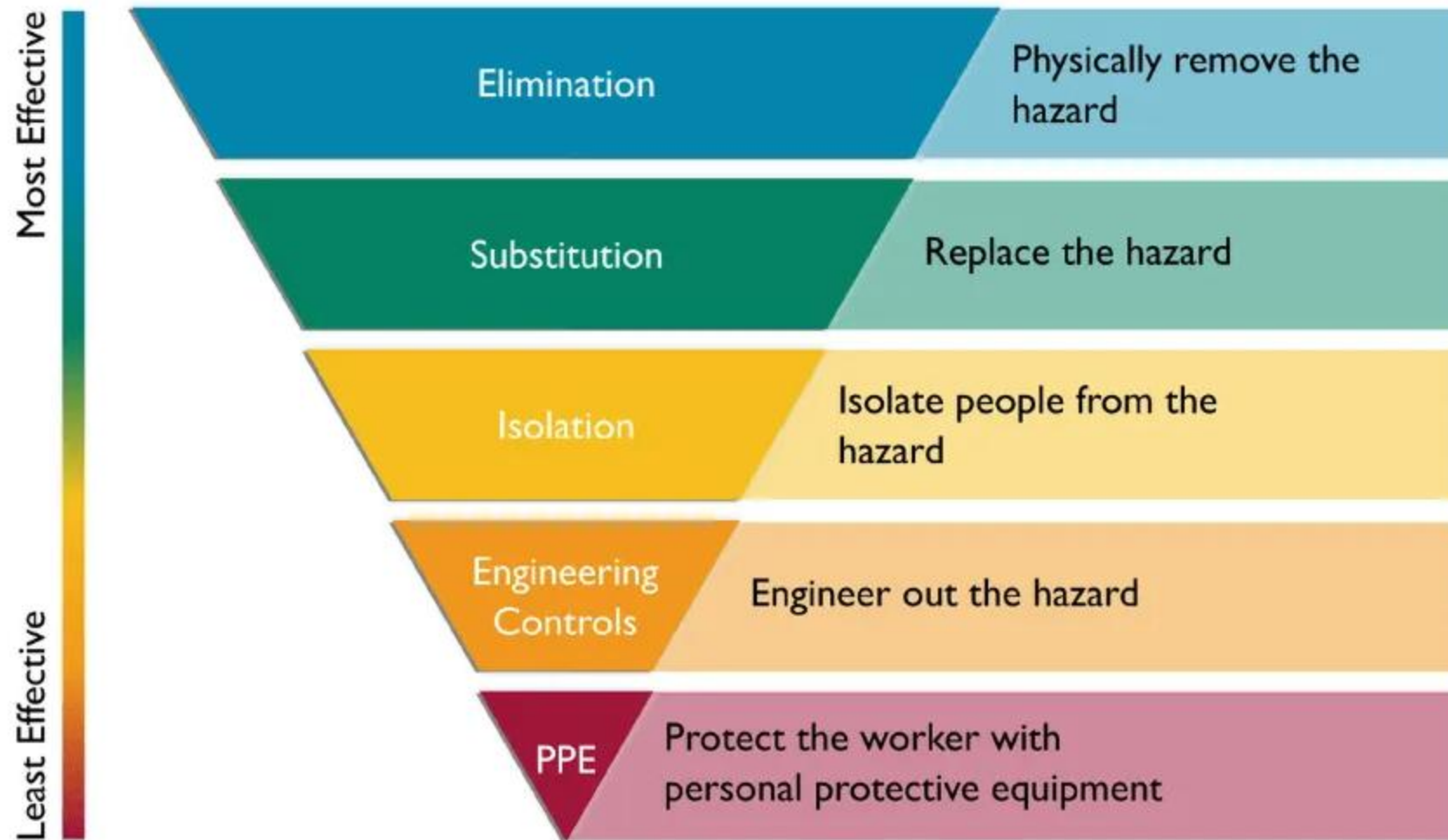
Major Accident Hazard (MAH):

- a) An event involving a fire, explosion, loss of well control or the release of a dangerous substance causing, or with a significant potential to cause, death or serious personal injury to persons on the facilities or engaged in an activity on or in connection with it;
- b) An event involving major damage to the structure of the installation or plant affixed to it or any loss in the stability of the installation causing, or with a significant potential to cause, death or serious personal injury to persons on the installation or engaged in an activity on or in connection with it;
- c) The failure of life support systems for diving operations in connection with the installation, the detachment of a diving bell used for such operations or the trapping of a diver in a diving bell or other subsea chamber used for such operations;
- d) Any other event arising from a work activity involving death or serious personal injury to five or more persons on the facilities or engaged in an activity on or in connection with it; or
- e) Any major environmental incident resulting from any event referred to in paragraph (a), (b) or (d) above

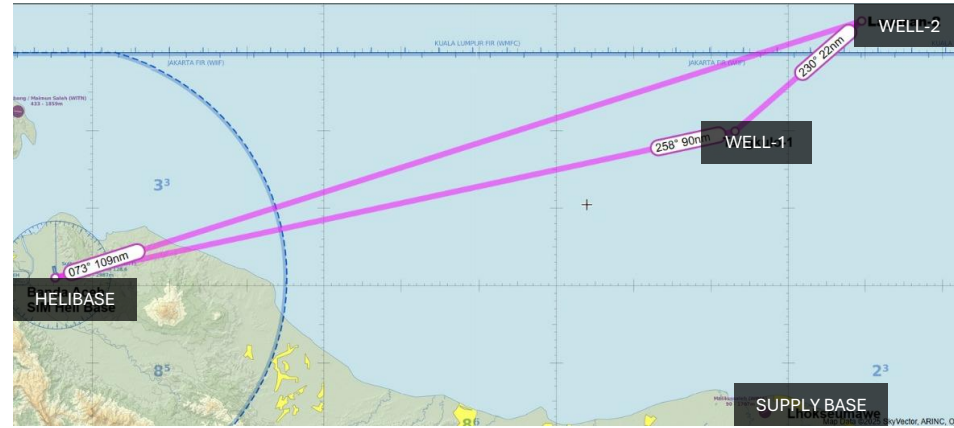


Can we work without “Risk” ?

Risk Control Hierarchy



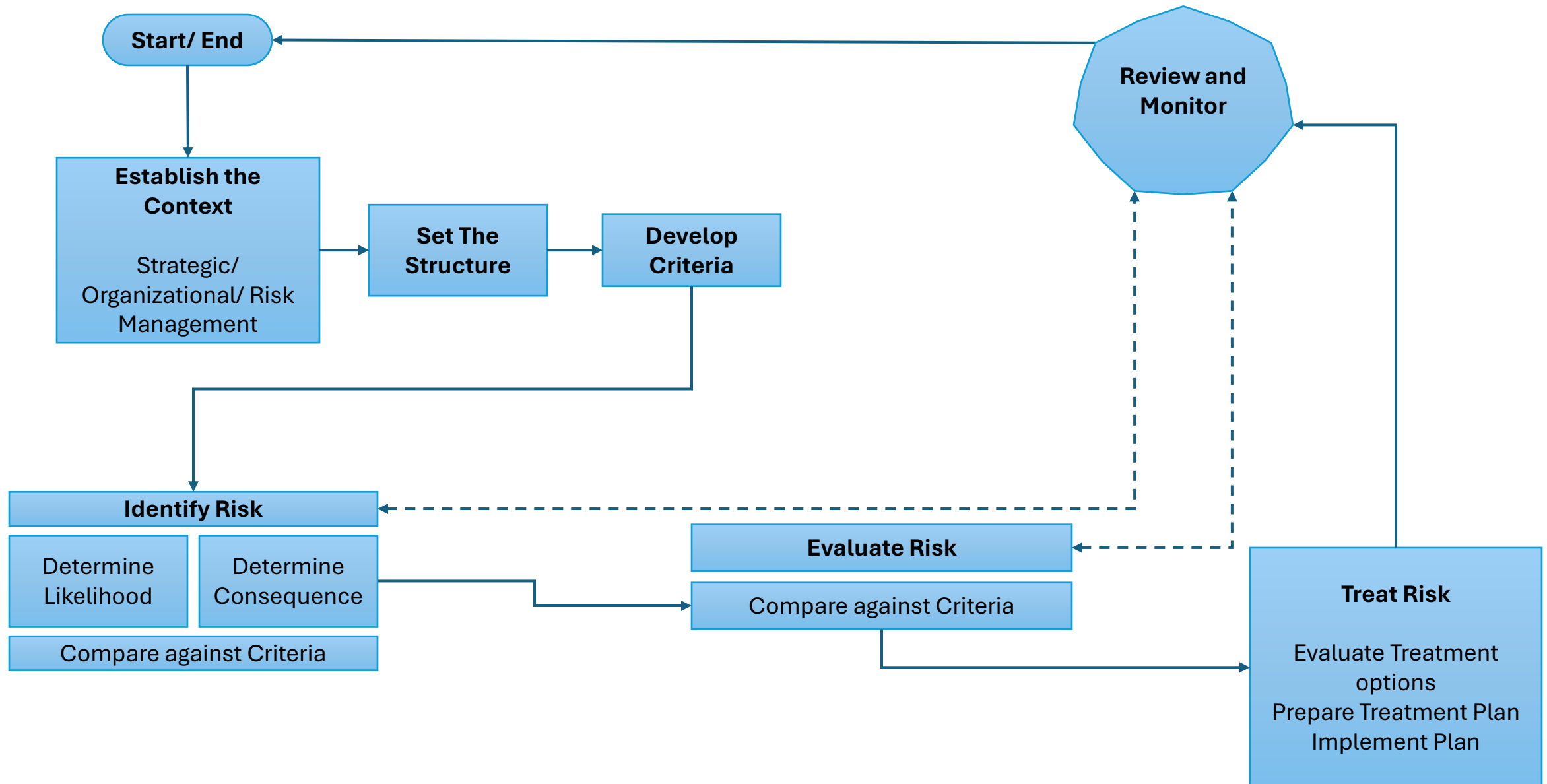
Identify The Hazard



- Exploration drilling
- Development drilling
- Deepwater drilling
- Remote area
- High technology equipment
- Tight operation window
- Personnel management



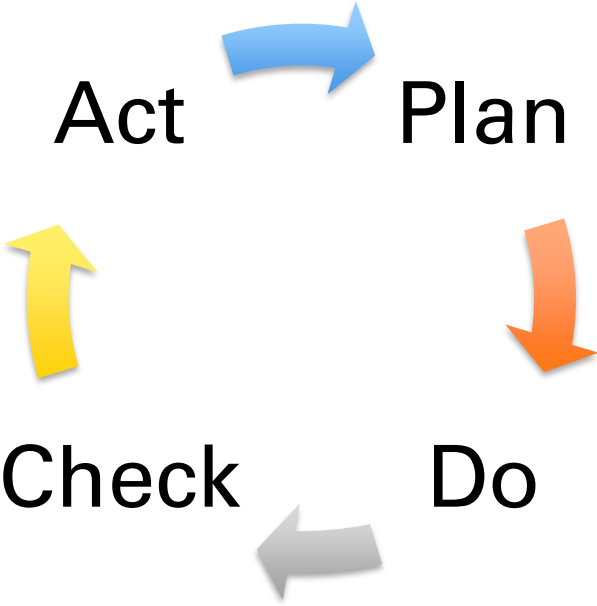
Risk Management Cycle



Risk Management Process

Hazard Identification			Initial Risk Assessment			Additional Control	Residual Risk Assessment		
Risk Description	Impact	Existing Control	Severity	Probability	Risk Level	Controls Needed	Severity	Probability	Risk Level

		Likelihood				
		Very Unlikely	Unlikely	Possible	Likely	Very Likely
Consequences	Catastrophic	MED	MED	HIGH	HIGH	HIGH
	Major	MED	MED	MED	HIGH	HIGH
	Moderate	LOW	MED	MED	MED	HIGH
	Minor	LOW	LOW	MED	MED	MED
	Negligible	LOW	LOW	LOW	MED	MED



Probability Matrix

Rating	Probability Description	Frequency in E&P Industry
Very Likely	The event is expected to occur in most circumstances. Probability $\geq 95\%$	Occurred more than 10 times during the past 15 years
Likely	The event is likely to occur in most circumstances. Probability $\geq 70\%$ but $< 95\%$	Occurred between 8 and 10 times during the past 15 years
Possible	There is a moderate chance the event will occur. Probability $\geq 35\%$ but $< 70\%$	Occurred between 4 and 8 times during the past 15 years
Unlikely	The event may most likely not occur. Probability $\geq 10\%$ but $< 35\%$	Occurred between 1 and 4 times during the past 15 years
Very Unlikely	The event may occur only in exceptional circumstances. Probability $< 10\%$	Occurred less than once during the past 15 years

Severity Matrix

Rating	Financial (USD)	HSSE	Legal, Compliance & Reputation	Organizational Capability
Catastrophic	>350m	Multiple fatalities / major environmental impact	Severe breach, license risk	Attrition >25%
Major	>20m–35m	Single fatality / serious LTI	Significant penalties	Attrition >20–25%
Moderate	>10m–20m	LTI / moderate impact	Moderate breach	Attrition >15–20%
Minor	>2.5m–10m	MTC / minor spill	Minor compliance breach	Attrition >5–15%
Negligible	≤2.5m	FAC / negligible impact	No material breach	Attrition ≤5%

*Best practice

- For High Level risk assessment often utilized whole criteria to get wider view of risk appetite
- For Operational risk assessment often utilized specific criteria to determine the risk control

Pro Tips...

Element	Definition	
Risk Level	Likelihood × Consequence	
Risk Categories	Low / Medium / High	
Risk Type	Inherent Risk (no controls) vs Residual Risk (with controls)	

Step	Description	Key Requirement
Hazard Identification	Identify hazards, environmental aspects, and quality issues	Consider lifecycle, past incidents, regulations
Risk Analysis	Determine potential consequences (HSSE, financial, compliance, reputation)	Include cause–effect relationship
Risk Assessment	Evaluate Likelihood × Consequence using risk matrix	Use data or professional judgment
Risk Classification	Categorize risk level	Low / Medium / High
Risk Evaluation Type	Assess both inherent and residual risk	With and without controls

Discussion